

```

GNU nano 2.2.6 File: /etc/hostapd/hostapd.conf Modified
interface=wlan0
driver=rtl8712drv
ssid=Onion Pi
hw_mode=g
channel=6
wpa_passphrase=Raspberry
wpa_key_mgmt=WPA-PSK
wpa_pairwise=TKIP
rsn_pairwise=CCMP

Get Help WriteOut Read File Prev Page Cut Text Cut For
Exit Chatty Where Is Near Page Undo Text To Spell

```

Figure 2: HostCon fig

We now need to change our IP routing tables so that connections via the Wi-Fi interface (wlan0) will be routed through the Tor software. Type the following commands to flush the old rules from the IP NAT table:

```

sudo iptables -F
sudo iptables -t nat -F

```

Type the following to route all DNS (UDP Port 53) from the interface wlan0 to the internal Port 53:

```

sudo iptables -t nat -A PREROUTING -i wlan0 -p udp --dport 53
-j REDIRECT --to-ports 53

```

Type the following to route all TCP traffic from the interface wlan0 to Port 9040:

```

sudo iptables -t nat -A PREROUTING -i wlan0 -p tcp --syn -j
REDIRECT --to-ports 9040

```

Next, check that the IP tables are right with the following command:

```

sudo iptables -t nat -L
Iptables.jpg

```

Now, if it all went well, save the IP tables to the old NAT file using the following command:

```

sudo sh -c "iptables-save > /etc/iptables.ipv4.nat"

```

It will automatically get loaded when the networking is set up on reboot. Next, let's create our log file, which is very handy for debugging, with the code given below:

```

sudo touch /var/log/tor/notices.log
sudo chown debian-tor /var/log/tor/notices.log
sudo chmod 644 /var/log/tor/notices.log

```

You can check it with the following command:

```

ls -l /var/log/tor

```

```

pi@raspberrypi:~$ sudo sh -c "iptables-save > /etc/iptables.ipv4.nat"
pi@raspberrypi:~$ cat /etc/iptables.ipv4.nat
# Generated by iptables-save v1.4.14 on Fri Jun 14 01:58:29 2013
*filter
:INPUT ACCEPT [1179:225338]
:FORWARD ACCEPT [1:67]
:OUTPUT ACCEPT [350:182821]
COMMIT
# Completed on Fri Jun 14 01:58:29 2013
# Generated by iptables-save v1.4.14 on Fri Jun 14 01:58:29 2013
*nat
:PREROUTING ACCEPT [97:11245]
:INPUT ACCEPT [34:754]
:OUTPUT ACCEPT [23:1900]
:POSTROUTING ACCEPT [24:1949]
-A PREROUTING -i wlan0 -p udp --dport 53 -j REDIRECT --to-ports 53
-A PREROUTING -i wlan0 -p tcp --tcp-flags FIN,SYN,RST,ACK SYN -j REDIRECT
--to-ports 9040
COMMIT
# Completed on Fri Jun 14 01:58:29 2013

```

Figure 3: IPv4 set-up

```

pi@raspberrypi ~$ sudo touch /var/log/tor/notices.log
pi@raspberrypi ~$ sudo chown debian-tor /var/log/tor/notices.log
pi@raspberrypi ~$ sudo chmod 644 /var/log/tor/notices.log
pi@raspberrypi ~$ ls -l /var/log/tor/*
-rw-r--r-- 1 debian-tor adm 2410 Jun 14 00:05 /var/log/tor/log
-rw-r--r-- 1 debian-tor adm 0 Jun 14 00:38 /var/log/tor/notices.log
pi@raspberrypi ~$ sudo service tor start
[ OK ] Starting tor daemon...done.
pi@raspberrypi ~$ sudo service tor status
ls -l tor is running.
pi@raspberrypi ~$ sudo update-rc.d tor enable
update-rc.d: using dependency based boot sequencing
pi@raspberrypi ~$

```

Figure 4: TOR status

To start the Tor service manually, use the following:

```

sudo service tor start

```

To check if Tor is actually running, use the command given below:

```

sudo service tor status

```

Finally, to make it start on boot, use the following command:

```

sudo update-rc.d tor enable

```

To test the anonymity of our Tor proxy, you could simply click on Onion Pi, which is available in your Wi-Fi list, and type the password you entered at the time of setting up the *hostapd* configuration file.

To check that the proxy is working, visit a website like <http://www.whatsmyip.org>, which will display your IP address as it sees it and also the matching domain name, if available. The IP address should not be from your Internet provider — in fact, if you reload the page, it should change! **END**

References

- [1] <https://www.raspberrypi.org/>
- [2] <https://github.com/nextgens/Tor>
- [3] https://en.wikipedia.org/wiki/Onion_routing

By: Srijan Agarwal

The author is a developer at WikiToLearn and an open source enthusiast. You can contact him at srijanagarwal.cse@gmail.com.